

TrES-3b

R-1296

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160426 · created 2026-07-31T16:47:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457504.9283 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.057
Transit depth (Rp/Rs)^2	3.08 +/- 2.0 [%]
Orbital Inclination (inc)	81.84 +/- 2.73
Airmass coefficient 1 (a1)	0.966 +/- 0.098
Airmass coefficient 2 (a2)	0.02 +/- 0.17
Scatter in the residuals of the lightcurve fit is	10.04 %
Best Comparison Star	#1 - [195, 40]
Optimal Aperture	1.9
Optimal Annulus	4.82
Transit Duration (day)	0.048 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.057 vs 0.16309 (deviation 0.23σ)
Duration fit vs published	0.048 d vs 0.0567 d (deviation 0.32σ)
Mid-time O–C	-3.9 min
Depth SNR	3.1
Residual scatter	10.04 %

Light curve

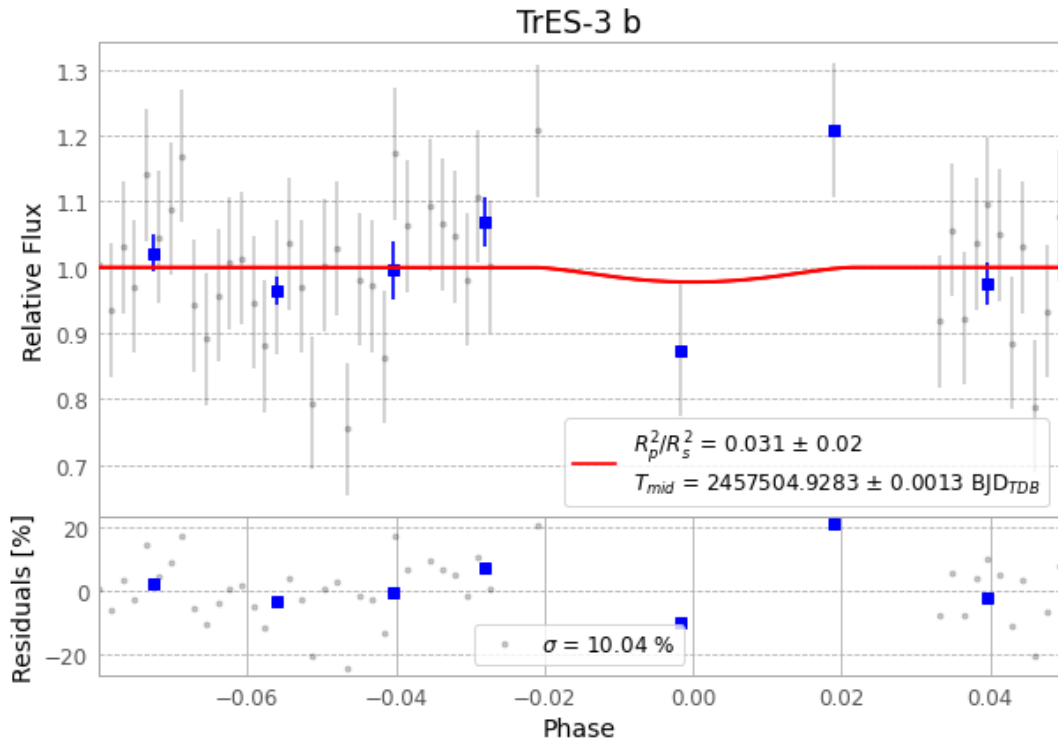


Figure 1 — FinalLightCurve_TrES-3 b_2016-04-26.png (SHA-256 ce102d422753a26f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 199], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 05959da56b87dbc3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), TRES-3160426074212.FITS → TRES-3160426114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160426.log (37821715b6fc...), FinalLightCurve_TrES-3 b_2016-04-26.png (ce102d422753...), FinalLightCurve_TrES-3 b_2016-04-26.csv (03bef3cee013...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)